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1 Implementation Plan: the SeisSol v3.4.1 RSSRW PREFERRED THERMAL CASE2 problem in SEAS-MFEM, with mixed flux

Date: 2026-07-08 Author: exploration + plan pass over seas-mfem-spatial-dyn-driver, branch system/spatial_dyn_driver

Successor to the v3.0.0 port plan (2026-06-19), in project_7.0_alternative/document/.

1.1 Overview

Run the **same physical problem** as the SeisSol deck

~/Downloads/seisol_quakeworx/safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2/

through seas_spatial_dyn_driver with MFEM's **mixed (adjacent-central) flux + explicit RK**, which SeisSol does not have. Same mesh, same CVM material, same Andersonian $k=1.8$ initial stress, same temperature-zoned rate-and-state + strong-rate-weakening friction, same 75 MPa / $R=2000$ m compact nucleation bump, same 150 s of simulated time.

This is **not** a bit-for-bit reproduction: SeisSol runs ADER-DG order 4 ($p=3$) with clustered LTS and a pure Godunov flux; MFEM runs $p=1$ global time stepping with a central flux on fault-adjacent faces. The *inputs* transfer exactly; the *discretization* is the experimental variable.

Headline

Four of the six input channels already work. The mesh is already in the repo in MFEM-readable form and is provably the same mesh SeisSol ran. Stress and material have existing .nc $\rightarrow$ data_projection_v1 HDF5 converters that need a re-run, not a rewrite. The flux/integrator combination, the FL=103 friction law, and the compact-bell nucleation are all implemented and unit-tested.

Two things genuinely block the run, and both are small:

#	Blocker	Fix size
B1	No way to feed a 3-D $rs_a(x,y,z)$ / $rs_{srW}(x,y,z)$ field to the friction resolver. The thermal deck's entire novelty (CTM temperature zoning) lives in that field.	~ 120 lines C++ + ~ 150 lines Python + 3 tests
B2	f_w default parser guard rejects $f_w = 0.0$, which is exactly what $RS_{\mu W} = 0.0$ in the deck asks for.	1 line + 1 test

Everything else is data preparation, config authoring, and one optional new output writer.

1.2 Background – verified findings

Every claim below was checked against the source or the files themselves during this exploration.

The deck (SeisSol side)

parameters.par + safs_fault.yaml + safs_initial_stress.yaml + safs_material_cvm.yaml:

Quantity	Value
Friction law	FL = 103 (rate-and-state slip law + strong rate weakening)
RS_{f0} / RS_{sr0}	$0.6 / 1e-6$
RS_b	0.019 (constant — v1.1.3 does not read spatial rs_b)
$RS_{\mu W}$ (f_w)	0.0
rs_{sl0} (Dc)	0.10 m (constant)
$RS_{iniSlipRate1/2}$	$1e-12 / 0.0$
rs_a, rs_{srW}	spatial , from safs_friction_thermal_case2.nc (ASAGI)
Nucleation	$Tnuc_s = 75e6 \cdot \exp(r^2/(r^2-R^2))$, $R = 2000$ m, centre (609062.8722, 3709528.1324, -10000.0), $Tnuc_n = Tnuc_d = 0$
Nucleation ramp	$s_0 = 0$, $t_0 = 1.0$ s (SeisSol smoothStep)
Fault reference	XRef, YRef, ZRef = (0, -1, 0), refPointMethod = 1 (reference vector)

Quantity	Value
Initial stress	safs_stress_andersonian_k1.8.nc, Andersonian C1 k=1.8, hydrostatic Pp already removed (effective), compression NEGATIVE
Material	safs_material_cvm.nc, compound {rho, mu, lambda}
Mesh	mesh_preferred.puml.h5, 1,241,419 tets / 250,588 vertices, 172,593 fault facets
Discretization	CFL = 0.5, ClusteredLTS = 2, order 4 (compile-time: SeisSol_Release_drome_4_elastic)
Flux	numflux = numfluxnearfault = 'godunov' (pure upwind everywhere)
EndTime	150.0 s
Plasticity	off

Sidcar grids (all ASAGI: NETCDF4, one compound var data, dims (z,y,x), float32 members, float64 strictly-increasing equidistant axes, z = elevation):

file	dims (z,y,x)	$\Delta x, \Delta y, \Delta z$ [m]	extent [m]	members
friction (CASE2)	107 × 194 × 263	1500, 1500, 200	x[303,696]k y[3612,3901.5]k z[−21000,+200]	rs_a ∈ [0.015, 0.0332]; rs_srW ∈ [0.05, 1000]
material (CVM)	194 × 361 × 452	1500, 1500, 250	x[28.5,705]k y[3543,4083]k z[−45000,+3250]	rho, mu, lambda
stress (Anders. k=1.8)	93 × 281 × 434	1000, 1000, 250	x[195,628]k y[3685,3965]k z[−20000,+3000]	s _{xx} ..s _{xz} ; s_{yz} ≡ s_{xz} ≡ 0

CASE2 friction profile (b = 0.019 constant, slope m = 8.0e−5 /°C):

$$\begin{aligned}
 a - b &= -0.004 & \text{for } T \leq 300 \text{ } ^\circ\text{C} & \quad (a = 0.015; \text{ VW everywhere} \\
 &\hookrightarrow \text{shallow}) \\
 a - b &= -0.004 + m \cdot (T - 300) & \text{for } T \geq 300 \text{ } ^\circ\text{C} & \quad (\text{zero-crossing exactly at} \\
 &\hookrightarrow 350 \text{ } ^\circ\text{C}) \\
 V_w &= 0.05 \text{ m/s} & \text{for } T \leq 350 \text{ } ^\circ\text{C} \\
 V_w &= 0.05 + (1000 - 0.05) \cdot (T - 350) / 50 & \text{for } 350 < T < 400 \text{ } ^\circ\text{C} \\
 V_w &= 1000 \text{ m/s} & \text{for } T \geq 400 \text{ } ^\circ\text{C}
 \end{aligned}$$

CASE1 (the sibling deck) adds a shallow VS cap at the 100 °C isotherm; **CASE2 has no shallow VS zone** — the shallow fault keeps $V_w = 0.05$ and can weaken dynamically. At the hypocentre column: 100 °C at 2.93 km, 350 °C at 13.19 km depth. $T(\text{hypocentre}) = 277.8 \text{ } ^\circ\text{C} \Rightarrow a = 0.015$, $V_w = 0.05$.

The mesh is already here, and it is the same mesh

```
$ shasum -a 256 .../meshing_deep19km/results/safv4_deep19km_sub500m_flattop_nwtrim_fj
↪ ixed_opt.puml.h5
    ~/Downloads/seisol_quakeworx/.../mesh_preferred.puml.h5
f08aef34... (identical)
```

The PUML the SeisSol job read was produced by `msh_to_puml.py` from

`safv4_deep19km_sub500m_flattop_nwtrim_fixed_opt_safstags.msh`

whose default map is 101 fault → 3, 102 top → 1, 103 bottom → 5, 104 sides → 5. That `.msh` is Gmsh **v2.2 ASCII** (MFEM's only supported flavour), 250,588 nodes, 1,543,222 elements (1,241,419 tets plus 301,803 boundary triangles), physical tags 101/102/103/104 + volume 1.

`seas_spatial_dyn_driver` with **no** [boundary] block falls back to exactly `fault_attr=101`, `natural={102}`, `absorbing={103,104}` (`spatial_dyn_driver.cpp:1070`). So the mesh channel is **zero work**.

For scale: the MFEM SAFS jobs already in production run `safv4_deep_500m_opt_safstags.msh` — **314,284 nodes / 1,846,898 elements**, i.e. *larger* than the deep19km mesh — on 2 nodes × 50 ranks.

What MFEM already implements

- **interior_flux="matrix" + mixed_flux="adjacent" + time_integrator="rk4"|"rk45"** is the intended heterogeneous-CVM mixed-flux mode. `BimaterialWaveOperator::SetMixedFluxMode` (`bimaterial_wave_operator.inl:1170`) explicitly *lifts* the old R-003 matrix×mixed-flux exclusion. Central flux is non-dissipative ⇒ **unstable under ADER**; both `ComputeMaxDt` overloads and a driver-level guard (`MatrixMixedFluxUnderAder`) abort on that combination. The comment at `wave_operator.hpp:335` claiming “mixed flux is scalar-only” is **stale**.
- **FL=103 equivalent:** `state_evolution = "slip_law_strong_rate_weakening"` (alias `"slip_law_srw"`), `slip_law_srw_psi.hpp: f_ss(V) = f_w + (f_LV - f_w)/(1+(V/V_w)8){1/8}`, `f_LV = max(0, f_0 - (b-a)·ln(V/V_0))`, `ψ_ss = a·ln((2V_0/V)·sinh(f_ss/a))`. Per-QP `a` and per-QP `V_w` are threaded through `Rate_SRW / SteadyState_SRW / PsiSS_SRW`.
- **Compact-bell nucleation:** `[nucleation] kind = "gradual overstress_compact_circular"` → `CompactBellFactor(r,R) = exp(r2/(r2-R2))`, strike-only (`tau2_nuc`), SCEC smooth-Step temporal ramp over `[0, T_nuc_s]` — the same function SeisSol uses for `t_0`. Parser at `spatial_friction.cpp:1363`, driver dispatch at `spatial_dyn_driver.cpp:1911`. **Both CLAUDE.md (§ Spatial driver nucleation mechanism) and spatial_friction_config_schema.md claim the spatial driver has only gradual overstress, and that rate-state ignores [nucleation]. Both statements are stale** — `tpv104/configs/tpv104_spatial.toml` is `law="rate_state" + kind="gradual overstress_compact_circular"` and is a validated benchmark. Fix both docs as part of this work.
- **Stress sidecar:** `[stress] kind="sidecar_hdf5"` → `StressField3D` (`stress_field_3d.hpp`), six `DataField3D` readers over a `data_projection_v1` HDF5 (`grid/{x,y,z}` float64 ascending, `fields/sigma_{xx,yy,zz,xy,yz,xz}` shaped `(Nx,Ny,Nz)`, `attrs` `schema_version`, `crs="EPSG:32611"`, `z_positive="elevation"`, per-dataset `min_value/max_value`). **Strict pass-through — compression POSITIVE**. The σ_{xy} negation at `spatial_dyn_driver.cpp:1716`

is in the ConstantTensor branch **only**; the sidecar branch does not touch signs.

- **Material sidecar:** [velocity] bundle reads fields/{Vp, Vs, density} from the same schema; far_field_clamp = true $\Rightarrow$ 00BPolicy::Clamp, the ASAGI nearest-edge hold. $\mu = \rho V_s^2$, $\lambda = \rho V_p^2 - 2\mu$. Evaluated per quadrature point (Mode::Coefficient), no P1 smearing.
- **Converters already written:**
 - seisol_quakeworx/toolbox/on_fault_stress_projection_csm/ csm_stress_nc_to_mfem_hdf5.py — ASAGI stress .nc $\rightarrow$ schema-v1 HDF5, applies the **global –1 sign flip** on all six components and already nudges max_value for identically-zero components (which s_yz/s_xz are here).
 - safs/project_7.0_preferred/velocity/code/build_pref_velocity_sidecar.py — safs_material_cvm.nc {rho,mu,lambda} $\rightarrow$ schema-v1 HDF5 {Vp,Vs,density}.
- **Checkpoint/restart**, ParaView fault output (7 fields: slip_{dip,strike}, slip_rate_{dip,strike}, traction_{dip,strike}, state_variable), free-surface slice.

Sign / frame conventions (established, audited 2026-07-07)

	MFEM	SeisSol	handoff
Frame	UTM 11N, x=E y=N z=Up	same	1:1
Stress	compression positive	compression negative	global $\times (-1)$ at the converter, once
Fault tangent frame	t1 = dip, t2 = strike (FaultBasis/Tandem)	1 = strike, 2 = dip	tau_strike_MFEM $\equiv$ Ts_SeisSol exactly (two flips cancel); tau_dip_MFEM = –Td_SeisSol
ref_normal	[0, –1, 0] (default)	XRef=(0, –1, 0), refPointMethod=1	identical
σ_n	> 0 compression	min(σ_n , 0)	—
Pore pressure	$\sigma_{n_eff} = \sigma_n - P_p$, applied unconditionally	—	nc is already effective $\Rightarrow P_{p_pa} = 0.0$ (else double-subtract)

The Andersonian nc's own global attribute states it: convention = "SeisSol compression-NEGATIVE; effective (hydrostatic P_p removed)".

1.3 Gap table

ID	Gap	Severity	Phase
G1	No 3-D field ingestion for <code>rs_a</code> / <code>rs_sr</code> . Resolver seeds <code>a,b</code> from scalars or a 1-D depth CSV; <code>v_w</code> from scalars, box rules, or a boxcar taper. <code>spatial_friction.cpp:2031</code> .	Blocker	3
G2	<code>MFEM_VERIFY(f_w_default > 0.0 && f_w_default < 1.0)</code> at <code>spatial_friction.cpp:713</code> rejects <code>f_w = 0</code> .	Blocker	2
G3	<code>velocity_safs.h5</code> was built from the <code>v3.0.0</code> CVM nc ($297 \times 196 \times 194$); the deck ships the statewide nc ($452 \times 361 \times 194$). <code>build_pref_velocity_sidecar.py</code> also hard-codes a path under the old <code>seisSol_quakeworx/</code> location, which no longer exists (relocated 2026-06-26).	Data	4
G4	No MFEM stress sidecar for <code>safs_stress_andersonian_k1.8.nc</code> .	Data	5
G5	No on-fault pickpoint / off-fault receiver time-series writer for SAFS. Station writers are hard-coded SCEC grids keyed on five fixed <code>[problem].tag</code> values. The deck ships 10 pickpoints + 10 receivers.	Feature	7 (opt.)
G6	No energy output (<code>SeisSol EnergyOutput=1</code>).	Cosmetic	not planned
G7	MFEM has no LTS . SeisSol's clustered LTS gave a $30.5 \times$ speedup over GTS on this mesh.	Accept	1 (mea- sure)
G8	MFEM <code>[mesh].order = 1</code> (p1) vs SeisSol order 4 (p3). Not the same accuracy per DOF	Decision	6
G9	<code>sigma_n_strength_floor_pa = 10e6</code> in every MFEM SAFS config. The Andersonian field has $\min \sigma_n = 2.15$ MPa on this fault, so the floor is active on shallow facets and makes them stronger than SeisSol. No SeisSol analog.	Decision	6
G10	Compact-bell radius: MFEM measures <code>r</code> in the fault plane (projection onto the per-DOF dip/strike basis, <code>spatial_nucleation.hpp:231</code>); SeisSol's Lua uses the full 3-D Euclidean distance. On a curved fault the two differ by the normal offset; MFEM's $r \leq$ SeisSol's r , so MFEM's bump is marginally wider.	Quantify	6
G11	<code>spatial_dyn_driver.cpp:1418</code> warns: <code>rate_state</code> at <code>np>1</code> uses end-of-step ψ on rank-seam fault QPs (1st order). Fires on every production SAFS run today.	Known	8

1.4 Constraints

1. **Byte-exact regression contract.** TPV102 / TPV104 / TPV205 / BP5 outputs must not change. Every addition below defaults to “absent $\Rightarrow$ old behaviour”. `make test` must stay green.
2. **Do not edit** the files `miniapps/seas/CLAUDE.md` lists as requiring extreme care:

<code>friction/dieterich_ruina.hpp</code>	<code>friction/state_evolution.hpp</code>
<code>domain/elasticity_operator.hpp</code>	<code>fault/fault_basis.hpp</code>

fault/rate_state_fault.hpp	solver/seas_operator.hpp
solver/time_stepper.hpp	config/bp5_params.hpp

Nothing in this plan needs to.

3. **No NetCDF in C++.** The driver links HDF5 only (data_field_3d.cpp uses the raw HDF5 C API). All .nc $\rightarrow$.h5 conversion stays in offline Python. Do not add a netcdf-c dependency.
4. **Gmsh v2.2 ASCII only** for [mesh].path.
5. **Frontera/Expanse runs need explicit user approval** before submission.
6. Mixed flux $\Rightarrow$ RK. mixed_flux != "none" under ADER aborts by design; do not "fix" that.

1.5 Phase 1 – Baseline measurement (no code)

Goal

Know the actual time-step and per-step cost on the deep19km mesh before committing to a wall-time budget.

Steps

1. Build the driver locally (conda activate mfem-dev; make seas_spatial_dyn_driver).
2. Author a throwaway config that points at the deep19km .msh with --no-sidecar-material (constant material), interior_flux="scalar", mixed_flux="none", time_integrator="ader", and run mpirun -np 8 ./seas_spatial_dyn_driver --config <tmp>.toml --dry-run --print-derived. Record h_min, dt_cfl.
3. Repeat with --time-integrator rk4 --mixed-flux adjacent (still constant material, scalar flux) and record dt_cfl.

Expected numbers (derive, then check)

h is the **inscribed diameter** $6 \cdot \text{Vol}/A_{\text{total}}$ (wave_operator.inl:96) — the same $2 \cdot r_{\text{insphere}}$ SeisSol uses. SeisSol reported $dt_{\text{min}} = 5.13722e-5$ s at order 4 ($N=3$), CFL 0.5, from $dt = cfl \cdot 2r / ((2N+1) \cdot c_p)$, hence $\min_e(h_e/c_{p,e}) = 7 \cdot 5.13722e-5 / 0.5 = 7.192e-4$ s.

MFEM: $dt = \text{MixedFluxCflFactor} \cdot (cfl \cdot \text{RkCflFactor}) \cdot \min_e(h_e/c_{p,e})$ where $\text{RkCflFactor} = 3/(2p+1) = 1.0$ at $p=1$ and $\text{MixedFluxCflFactor}(\text{Adjacent}, \text{rk_aware}) = 0.6$ (wave_operator.hpp:870; spatial_friction.hpp:711; spatial_dyn_driver.cpp:2221):

$dt(p=1, \text{rk4}, \text{adjacent}) = 0.6 \cdot 0.5 \cdot 7.192e-4 = 2.16e-4$ s $\rightarrow \sim 695,000$ steps $\times 4$
 $\hookrightarrow$ RHS for 150 s

$dt(p=1, \text{ader}, \text{upwind}, cfl_{\text{dg_safety}}=3) = (0.5/9) \cdot 7.192e-4 = 4.00e-5$ s $\rightarrow \sim 3.75$ M
 $\hookrightarrow$ steps

$dt(p=1, \text{ader}, \text{upwind}, cfl_{\text{dg_safety}}=1) = (0.5/3) \cdot 7.192e-4 = 1.20e-4$ s $\rightarrow \sim 1.25$ M
 $\hookrightarrow$ steps

Note `--cfl-dg-safety` is **inert on the RK path** — it only scales `CflSafetyFactor`, which the ADER branch uses. The mixed-flux+RK production target gets no benefit from Lever A.

For calibration: SeisSol reached only $t \approx 98$ s of 150 s in 300 min on 160 ranks (job hit the Tapis

time limit), *with* a $30.5\times$ LTS speedup. Expect MFEM GTS at p1 to need a **checkpoint/restart chain**, not a single 2 h job.

Acceptance

dt_cfl printed and within $2\times$ of the estimate above. If it is wildly smaller, a sliver tet is driving h_min and that must be investigated before anything else (see project_flattop_badtet_fix / QUALITY_REPORT_deep19km.md, which reports eta_min PASS).

1.6 Phase 2 – Allow $f_w = 0$ (G2)

Goal

RS_muW = 0.0 (zero residual friction — the rate-state analogue of LSW $\mu_d = 0$) must parse.

Why it is safe

f_w enters only additively: $f_{ss} = f_w + (f_{LV} - f_w)/(1+(V/V_w)^8)^{1/8}$. There is no division by f_w and no $\log(f_w)$. $\psi_{ss} = a \cdot \ln((2V_0/V) \cdot \sinh(f_{ss}/a))$ needs $f_{ss} > 0$; with $f_w = 0$, $f_{ss} = f_{LV}/(1+(V/V_w)^8)^{1/8}$ and $f_{LV} = \max(0, f_0 - (b-a) \cdot \ln(V/V_0))$ only reaches 0 at $V = V_0 \cdot \exp(f_0/(b-a)) = 1e-6 \cdot e^{150}$, which is unreachable. $f_{ss} > 0$ strictly $\Rightarrow \psi_{ss}$ finite. SeisSol runs this deck with RS_muW = 0 on v1.1.3. The deck header flags it as an “energy-max idealization below Lachenbruch & Sass $\mu_d \approx 0.10$ ”.

Files to change

- spatial_friction.cpp:713 — relax to $f_w_default \geq 0.0$ & $f_w_default < 1.0$; update the R-004 comment to record *why* 0 is admissible and that $V_w > 0$ is the load-bearing guard (never $V_w = 0$ under FL=103).

Files to create

- seas_test_slip_law_srw_fw_zero/ (or extend seas_test_slip_law_srw_psi): assert PsiSS_SRW(V , V_w , a) is finite and monotone for $f_w = 0$ across $V \in [1e-12, 10]$ m/s with $a=0.015$, $b=0.019$, $f_0=0.6$, $V_0=1e-6$, $V_w=0.05$; assert the parser accepts $f_w_default = 0.0$ and still rejects -0.1 and 1.0 .

Acceptance

- make test green.
 - ParseSpatialFrictionConfigString with $f_w_default = 0.0$ + state_evolution = "slip_law_srw" returns without abort.
-

1.7 Phase 3 – 3-D friction-field sidecar (G1) – the only real feature work

Goal

Let [friction.rate_state] seed per-DOF a and V_w from a data_projection_v1 HDF5, exactly as [stress] and [velocity] already do. This is what carries the CTM temperature zoning.

Design

Mirror the stress/velocity pattern. Do **not** invent a new file format; reuse DataField3D (trilinear, 00BPolicy::Clamp = ASAGI edge-hold).

Why baked a/V_w and not a temperature field + inline $a(T)$ formula: SeisSol itself reads baked rs_a/rs_srW through ASAGI. Reading the same two fields means the two codes sample *identical* friction, and the $a(T)/V_w(T)$ piecewise logic stays in one place (build_friction_nc_thermal.py) instead of being duplicated in TOML.

1.7.2.1 Python: safs/project_7.0_preferred/friction/code/build_pref_friction_sidecar.py (new) Decision D5 (user, 2026-07-08): MFEM never reads .nc. The friction sidecar is the same file type as the velocity sidecar — a data_projection_v1 HDF5, written by the **same shared helper** safs/project_7.0_alternative/velocity/code/sidecar.py::write_sidecar that build_pref_velocity_sidecar.py already uses. Do **not** hand-roll an HDF5 writer (the stress converter has its own inline copy; that is technical debt, not a template).

So this script is a sibling of build_pref_velocity_sidecar.py, not of csm_stress_nc_to_mfem_hdf5.py:

```
# in : safs_friction_thermal_case2.nc    compound data{rs_a, rs_srW}, dims (z,y,x), f4
# out: ../friction/results/thermal_case2_mfem/friction_safs.h5
sys.path.insert(0, f"{ALT}/velocity/code")
from sidecar import write_sidecar

data = netCDF4.Dataset(NC).variables["data"][:]          # (z, y, x) compound
rs_a  = np.transpose(np.asarray(data["rs_a"],    np.float64), (2, 1, 0))  # -> (x,
    ↪ y, z)
rs_srW = np.transpose(np.asarray(data["rs_srW"], np.float64), (2, 1, 0))

write_sidecar(
    OUT, x, y, z,
    fields      = {"rs_a": rs_a, "rs_srW": rs_srW},
    attrs       = {"schema_version": "data_projection_v1", "crs": "EPSG:32611",
                  "units": "m", "z_positive": "elevation",
                  "source": "safs_friction_thermal_case2.nc (CTM shinevar2024, CASE2
    ↪ "
                  "a(T)/V_w(T)) -> build_pref_friction_sidecar.py",
                  "mesh_tag": "safv4_deep19km_sub500m_flattop_nwtrim_fixed_opt"},
    field_bounds = {"rs_a": (0.010, 0.050, "1"),
                  "rs_srW": (0.05, 1000.0, "m/s")},
)
```

write_sidecar already enforces: strictly-monotone axes, no NaN, shape (Nx,Ny,Nz), and every cell inside field_bounds — i.e. it gives the build script's G2 gate ($rs_a > 0$, $rs_srW \in [0.05, 1000]$) for free and raises rather than warns.

Add one guard the helper cannot know about: `assert a(hypocentre) == 0.015` and `V_w(hypocentre) == 0.05` to `1e-9` by trilinear sampling before writing (the deck's own G4 gate).

Resulting layout, byte-comparable to velocity_safs.h5:

```
grid/x (263,) f8  grid/y (194,) f8  grid/z (107,) f8      # ascending, equidistant
fields/rs_a  (263,194,107) f8  attrs units="1",  min_value, max_value
fields/rs_srW (263,194,107) f8  attrs units="m/s", min_value, max_value
```

1.7.2.2 C++: schema spatial_friction.hpp:

```
/// [friction.rate_state.sidecar] – per-DOF a / V_w (and optionally b, Dc) from a
/// data_projection_v1 HDF5. Absent ⇒ enabled=false ⇒ resolver behaviour
⇒ byte-identical.
struct FrictionSidecarSpec
{
    bool          enabled          = false;
    std::string path;              // required when enabled
    std::string a_field            = "rs_a";
    std::string V_w_field          = "rs_srW";
    std::string b_field;           // "" ⇒ keep b_default
    std::string Dc_field;          // "" ⇒ keep Dc_default
    bool          far_field_clamp = true;  // 00BPolicy::Clamp (ASAGI edge hold)
};
```

Add FrictionSidecarSpec sidecar; to RateStateBlock.

Loaded fields (owned by the driver, handed to the resolver):

```
struct RateStateSidecarFields
{
    std::unique_ptr<DataField3D> a, V_w, b, Dc;  // b/Dc may be null
};
```

SpatialFrictionResolver gains an optional ctor argument `std::shared_ptr<const RateStateSidecarFields>`; the default ctor leaves it null, so every existing call site (and every TPV/BP5 test) is byte-unchanged.

1.7.2.3 C++: parser (spatial_friction.cpp, in parse_rate_state)

- Read [friction.rate_state.sidecar]; path required and non-empty when the table is present.
- `MFEM_VERIFY(!(sidecar.enabled && depth_profile.enabled))` — both seed a/b; a config that sets both is ambiguous and must abort with a message naming both keys.
- Keep the existing `a_default < b_default` guard satisfied by the config (`0.015 < 0.019`); `a_default` becomes a pure fallback.

1.7.2.4 C++: resolver (resolve_rs_impl, around spatial_friction.cpp:2031) Insert **after** the depth-profile/scalar seed and **before** the spatial-rule loop, so box / boxcar_taper rules can still override a sidecar value:

```

if (sidecar_) // null  $\Rightarrow$  untouched legacy path
{
    a_i = sidecar_>a ->Evaluate(x, y, z);
    V_w_i = sidecar_>V_w->Evaluate(x, y, z);
    if (sidecar_>b) { b_i = sidecar_>b ->Evaluate(x, y, z); }
    if (sidecar_>Dc) { Dc_i = sidecar_>Dc->Evaluate(x, y, z); }
}

```

The existing $V_w_i > 0$ guard (:2166) and the per-DOF $a > b$ allowance (R-011, :2153) already cover the VS zone (a up to $0.0332 > b = 0.019$).

1.7.2.5 C++: driver (spatial_dyn_driver.cpp)

- Construct RateStateSidecarFields right before ResolveRateState (after the mesh + material exist).
- OOBPolicy = Clamp when far_field_clamp, else Abort.
- Extend --print-derived to report, over fault DOFs: min/max/mean of a and V_w , the count of DOFs with $a - b < 0$ (the VW area), and the values at [hypocenter]. This is the offline $\leftrightarrow$ online cross-check against the SeisSol build script's own G4 gate.

Files to create

- safs/project_7.0_preferred/friction/code/build_pref_friction_sidecar.py
- seas_test_spatial_friction_sidecar/ — three tests:
 1. **Round-trip**: write a synthetic $3 \times 3 \times 3$ sidecar with a known linear $a(x, y, z)$; resolve at 8 corners and the centre; assert trilinear values to $1e-12$.
 2. **Clamp**: query outside the hull with far_field_clamp = true $\Rightarrow$ edge value; with false $\Rightarrow$ abort.
 3. **Rule precedence**: sidecar + a box rule setting $V_w \Rightarrow$ rule wins inside the box, sidecar outside.
- Parser tests: sidecar + depth_profile together $\Rightarrow$ abort; missing path $\Rightarrow$ abort.

Files to change

- spatial_friction.hpp (schema + resolver ctor)
- spatial_friction.cpp (parse + resolve)
- spatial_dyn_driver.cpp (load + --print-derived)
- safs/project_7.0_alternative/document/spatial_friction_config_schema.md (document the new block; also fix the stale “rate-state ignores [nucleation]” claim at lines 256-258 / 376-377)
- miniapps/seas/CLAUDE.md (fix the stale “single nucleation kind” paragraph)

Acceptance

- make test green; seas_test_spatial_friction_resolver byte-unchanged (no sidecar $\Rightarrow$ no behaviour change).
- --print-derived on the real sidecar reports $a(\text{hypo}) = 0.015$, $V_w(\text{hypo}) = 0.05$, min $a = 0.015$, max $a \approx 0.0332$, min $V_w = 0.05$, max $V_w = 1000$.

- Fault-DOF VW fraction is consistent with “350 °C isotherm at ~13.2 km at the hypocentre column”.

1.8 Phase 4 – Rebuild the CVM material sidecar (G3)

Goal

velocity_safs.h5 sampled from the **same** safs_material_cvm.nc the SeisSol job read.

Files to change

- safs/project_7.0_preferred/velocity/code/build_pref_velocity_sidecar.py
 - Replace the hard-coded source path

```
NC = f"{ALT}"
↪ /seisol_quakeworx/safs_seisol_v3_0_0_LSW_PREFERRED/safs_material_cvm.nc"
```

(this path **no longer exists** — seisol_quakeworx moved to safs/seisol_quakeworx/ on 2026-06-26) with --in / --out argparse options. Keep the old defaults out; make both required.

Output

project_7.0_preferred/velocity/results/multiscale_statewise_cvm_v3_4_1/velocity_safs.h5 (new directory so the v3.0.0 sidecar is not clobbered; both are gitignored, scp to Expanse).

Source nc: safs_material_cvm.nc from the deck directory (452 × 361 × 194; the existing h5 is 297 × 196 × 194 — a **different, smaller** grid).

Acceptance

- $v_p = \sqrt{(\lambda + 2\mu)/\rho}$, $v_s = \sqrt{\mu/\rho}$, density = ρ , all finite, $\rho > 0$, $\mu \geq 0$.
- Grid attrs: crs="EPSG:32611", z_positive="elevation", axes ascending.
- Mesh bbox x[253447,711231] y[3532686,3956757] z[−39329,+26] pokes outside the CVM hull only at the far-field absorbing corners (x high ≤ 6.2 km, y low ≤ 10.3 km) $\Rightarrow$ **far_field_clamp = true is mandatory**, exactly as SeisSol relies on ASAGI’s edge clamp + the yaml !ConstantMap.

1.9 Phase 5 – Convert the Andersonian stress field (G4)

Goal

safs_stress_andersonian_k1.8.nc $\rightarrow$ schema-v1 HDF5, compression-**positive**, effective Pa.

Command

```
python3 safs/seisol_quakeworx/toolbox/on_fault_stress_projection_csm/csm_stress_nc_
↪ to_mfem_hdf5.py \
--in ~/Downloads/seisol_quakeworx/safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE_
↪ 2/safs_stress_andersonian_k1.8.nc \
```

```
--out safs/project_7.0_preferred/stress/results/andersonian_k1p8_mfem/safs_stress_
↪ andersonian_k1.8_mfem.h5
```

The script needs no modification: the member names are already `s_xx..s_xz`, and its `min<max` nudge already covers `s_yz` $\equiv$ `s_xz` $\equiv$ 0.

Cautions

- The script's `HYP0_DEFAULT = (606971.0, 3707270.0, -4965.62)` is the **v3.0.0** hypocentre used only for a diagnostic print. Pass the v3.4.1 on-fault hypocentre (609062.8722, 3709528.1324, -10000.0) if the script exposes a flag; otherwise ignore the print (it is not used in the output).
- **[pore_pressure] P_p_pa = 0.0**. The `nc` is already effective (its own convention attribute says so). A non-zero `P_p_pa` would double-subtract in `ProjectFaultPreStress`.
- Grid `z[-20000,+3000]` and `x[195,628]k / y[3685,3965]k` contain the fault bbox `x[363.5,626.7]k y[3692.0,3839.0]k z[-19329,+26]`, so the default `00BPolicy::Abort` in `ApplyCsmStress-Sidecar` is fine — stress is only projected onto **fault** DOFs.

Acceptance

- Sampled at the hypocentre facet: $\sigma_n = 169.83$ MPa, $\mu_{app} \leq 0.2931$, $\tau_0 \geq 0.006$ MPa — the numbers the deck header records for this fault.
- $\min \sigma_n$ over fault DOFs ≈ 2.15 MPa, 0 non-finite.

1.10 Phase 6 – Production TOML

File to create

```
safs/project_7.0_preferred/config/
  spatial_friction_safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2.toml

[meta]
schema_version = 1
law             = "rate_state"
description     = "MFEM port of
↪ seisol_quakeworx/safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2 (FL=103, CTM
↪ temperature-zoned rs_a/rs_srW via sidecar, f_w=0, Andersonian k=1.8 effective
↪ stress, statewide CVM) with MIXED FLUX (adjacent) + RK. 'Do what SeisSol did, but
↪ with mixed flux.'"

[material_constant_fallback]                # only reachable via --no-sidecar-material
lambda = 32.0e9
mu      = 32.0e9
rho     = 2670.0

[pore_pressure]
P_p_pa      = 0.0                        # nc is EFFECTIVE (hydrostatic Pp already
↪ removed)
P_p_grad_pa_per_m = 0.0
min_sigma_n_pa   = 1.0e6                # inert: min sigma_n on this fault is 2.15 MPa
```

```

[mesh]
path = "safes/project_7.0_preferred/meshing_deep19km/results/safv4_deep19km_sub500m_
↳ flattop_nwtrim_fixed_opt_safstags.msh"
order = 1                                # DECISION D2 (SeisSol runs p3)

[velocity]
use_sidecar      = true
model            = "multiscale_statewise"
dataset_root     = "safes/project_7.0_preferred"
override_path    = "safes/project_7.0_preferred/velocity/results/multiscale_statewise_
↳ cvm_v3_4_1/velocity_safes.h5"
far_field_clamp = true                    # MANDATORY: mesh far field exceeds CVM hull

[material]
kind             = "sidecar_hdf5"         # required: interior_flux="matrix" forbids
↳ kind="constant"
sidecar_path     = "safes/project_7.0_preferred/velocity/results/multiscale_statewise_cvm_
↳ _v3_4_1/velocity_safes.h5"

[stress]
kind             = "sidecar_hdf5"
sidecar_path     = "safes/project_7.0_preferred/stress/results/andersonian_k1p8_mfem/safes_
↳ _stress_andersonian_k1.8_mfem.h5"

[numerics]
ader_order       = 2                      # ignored on the RK path
mixed_flux       = "adjacent"
interior_flux    = "matrix"               # heterogeneous CVM ⇒ bimaterial Riemann
time_integrator  = "rk4"                  # central flux is UNSTABLE under ADER
cfl              = 0.5                     # = SeisSol CFL
use_pml          = false

[time]
tfinal           = "150s"                  # = SeisSol EndTime
t_initial        = 0.0
dt_initial       = "auto"
dt_max           = "0.1s"                  # = SeisSol FixTimeStep

[output]
output_dir       = "output_safes_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2"
restart_prefix   = "cp"
paraview_volume  = "off"
paraview_bulk    = "off"
paraview_fault   = "hdf5"

```

```

paraview_fault_dt      = "4.0s"          # = SeisSol Elementwise printtimeinterval_sec
paraview_fault_zfp_tol = 1.0e-12
paraview_free_surface  = "vtu"
paraview_free_surface_dt = "4.0s"        # = SeisSol SurfaceOutputInterval
max_snapshots          = 5000
checkpoint_every_steps  = 10000

```

```
[nucleation]
```

```
kind = "gradual overstress_compact_circular"
```

```
[nucleation.gradual overstress_compact_circular]
```

```

# = SeisSol Tnuc_s = 75e6 * exp(r^2/(r^2-R^2)), R=2000, smoothStep over [s_0,
  ↪ s_0+t_0] = [0,1] s.
# Centre = the on-fault point at EXACTLY 10 km depth closest to the deck-lineage
  ↪ geographic target.
center_x_m   = 609062.8722
center_y_m   = 3709528.1324
center_z_m   = -10000.0
radius_m     = 2000.0
delta_tau_pa = 75.0e6
T_nuc_s      = "1.0s"

```

```
[friction]
```

```
sigma_n_strength_floor_pa = -1.0          # DECISION D1: -1 = disabled = SeisSol parity
```

```
[friction.rate_state]
```

```

state_evolution = "slip_law_strong_rate_weakening" # = SeisSol FL=103
f_w_default     = 0.0          # = RS_muW (requires Phase 2)
V_w_default     = 0.05         # FALLBACK ONLY (sidecar governs)
a_default       = 0.015        # FALLBACK ONLY (sidecar governs); must stay < b_default
b_default       = 0.019        # = RS_b (constant; v1.1.3 reads no spatial rs_b)
Dc_default      = 0.10         # = rs_sl0 (constant)
V_0_default     = 1.0e-6       # = RS_sr0
f_0_default     = 0.6          # = RS_f0
V_init_default  = 1.0e-12      # = RS_initSlipRate1
sigma_n_default = 169.83e6     # FALLBACK ONLY (live sigma_n is per-DOF from the stress
  ↪ sidecar)
eta             = "auto"       # 0.5*sqrt(mu*rho) per DOF == SeisSol radiation damping

```

```
[friction.rate_state.sidecar]
```

```
# NEW (Phase 3) – same file type as
```

```
↪ [velocity]
```

```

path           = "safs/project_7.0_preferred/friction/results/thermal_case2_mfem/fric_
  ↪ tion_safs.h5"
a_field        = "rs_a"
V_w_field      = "rs_srW"

```

```

far_field_clamp = true                                # CTM hull covers the fault, not the far field

# No [boundary] block: the driver falls back to fault=101, natural={102},
  ↪ absorbing={103,104},
#   which is exactly what the safstags mesh uses.
# No [fault_geometry] block: defaults ref_normal=[0,-1,0], up=[0,0,1] ==
  ↪ XRef/refPointMethod=1.
# No [friction.rate_state.depth_profile]: mutually exclusive with [.sidecar].

```

Decisions – RESOLVED (user, 2026-07-08)

D1 — σ_n strength floor pa: two arms. Every existing MFEM SAFS RS config sets $10.0e6$. On this fault $\min \sigma_n = 2.15$ MPa, so the floor is **active** on shallow facets and makes them stronger than SeisSol's. Author **both** configs:

config	σ_n strength floor pa	role
..._THERMAL_CASE2.toml	-1.0 (disabled)	parity arm — run this first fallback arm, only if the parity arm shows trace-cell runaway
..._THERMAL_CASE2_snfloor10.t	$10.0e6$	

The two files differ in **that one key and output_dir** and nothing else. Runaway signature (from project_pre_k25_case2_runaway_slip_not_mesh): isolated shallow cells at σ_n 0.1–5 MPa sliding ~ 4 m/s indefinitely, visible in the first few fault snapshots. $\min \sigma_n$ pa = $1.0e6$ is a *different* knob (it clamps σ_n eff, not the strength) and is inert here — keep it in both.

D2 — [mesh].order: p1 production + p2 convergence window. order = 1 for the 150 s run (matches every existing SAFS production run and the Phase-1 cost estimate). Separately run order = 2 with tfinal = "20s" as a convergence check: if the p2 rupture front and peak slip rate agree with p1 to within the snapshot cadence, p1 resolves the cohesive zone well enough; if not, the p1 result is quantitatively suspect and must be reported as such. Under mixed-flux+RK, p2's dt penalty is only $RkCflFactor(2)/RkCflFactor(1) = 0.6$, but DOFs rise $\sim 2.5 \times \Rightarrow \sim 4 \times$ total cost. Add this as gate 6b in Phase 8.

D3 — time_integrator: rk4. Fewer RHS evaluations per step than rk45 at the same fixed dt. The existing sbatch scripts pass --time-integrator rk45; the sbatch override wins, so set the default in the sbatch to rk4 for this case.

D4 — tfinal: 150 s (= EndTime). But the reference SeisSol run only reached $t \approx 98$ s before its Tapis time limit, so **0–98 s is the entire comparison window** the reference data supports.

D5 — friction sidecar file type: data_projection_v1 HDF5, identical in kind to the velocity sidecar, written by the shared sidecar.py::write_sidecar. MFEM reads no NetCDF anywhere. See Phase 3.

D6 — receivers/pickpoints (Phase 7): deferred to a follow-up. Phases 1–6 + 8 give a runnable, comparable simulation on fault snapshots and free-surface slices; the per-point .dat writers upgrade that to waveform misfit once there is a run worth comparing.

1.11 Phase 7 – SAFS receiver / pickpoint output (G5) – optional, scope decision

Without this, the only comparison surface is ParaView fault snapshots at 4 s cadence and free-surface slices. The deck's `safs-faultreceiver-*.dat` (10 on-fault, `printtimeinterval = 10` steps) and `safs-receiver-*.dat` (10 off-fault, `pickdt = 0.005 s = 100 Hz`) are the highest-signal comparison data and they already exist for the SeisSol side.

Design

Generalize the existing `wire_stations` block (`spatial_dyn_driver.cpp:3297`) rather than adding a fifth hard-coded SCEC grid:

- New TOML: `[stations] on_fault_file = "...", off_fault_file = "...", on_fault_dt, off_fault_dt.`
- Reuse the `.dat` format the deck ships (whitespace-separated `x y z`, one point per line — see `safs_pickpoints.dat` / `safs_receivers.dat`, both 10 lines).
- On-fault: nearest fault DOF per point (assert the residual is $< 0.5 \cdot \text{median fault edge} = 200 \text{ m}$); write `t`, `slip_strike`, `slip_dip`, `slip_rate_strike`, `slip_rate_dip`, `traction_strike`, `traction_dip`, `sigma_n`, `psi`.
- Off-fault: MFEM point-location on the volume mesh; write `t`, `vx`, `vy`, `vz`. Receivers sit at `z = -1 m` precisely because SeisSol v1.1.3 drops `z = 0` points; MFEM has no such restriction, but keep `z = -1` so both codes sample the same location.

Recommendation

Defer to a follow-up. Phases 1–6 + 8 produce a runnable, comparable simulation. Phase 7 upgrades the comparison from qualitative (rupture front, slip map) to quantitative (waveform misfit). Do it once the first run has produced something worth comparing.

1.12 Phase 8 – Job scripts and validation

Files to create

```
jobs/safs/safs_expanse/  
  spatial_dyn_ratestate_v3_4_1_pref_thermal_case2_cvm_mixedflux_rk_deep19km_expanse_  
  ↪ 2N100r_compute_2hr_safs.sbatch  
  ..._2hr_restart_safs.sbatch          # the --restart sibling
```

Copy the existing `..._v3pref_vw002_cvm_mixedflux_rk_safv4deep..._2hr_safs.sbatch` verbatim and change only `SAFS_CONFIG`, `SAFS_MESH`, `SAFS_OUT`, and the job name. Keep the existing `mixed_flux != none && integrator == ader abort guard`.

Promote the validated perf levers: `--deriv-cache` is $\leq 1e-12$ -parity and works on the RK path (`--face-cache` and `--shared-ck-recursion` are ADER-only; `--cfl-dg-safety` is inert on RK). Today's production scripts pass none of them.

Sidecars to scp to Expanse alongside the mesh (all gitignored, ~700 MB total): `velocity_safs.h5` (from Phase 4), `safs_stress_andersonian_k1.8_mfem.h5` (Phase 5), `safs_friction_thermal_case2_mfem.h5` (Phase 3).

Validation gates, in order

1. **--dry-run --print-derived, np=1, local.** Confirms mesh reads, all three sidecars load, boundary attrs resolve to 101/102/103/104, dt_cfl matches Phase 1.
2. **--print-derived friction gate.** $a(hypo)=0.015$, $V_w(hypo)=0.05$, $\min/\max a = 0.015/0.0332$, $\min/\max V_w = 0.05/1000$. Compare against `build_friction_nc_thermal.py`'s G2/G4 gates.
3. **Stress gate.** $\sigma_n(hypo \text{ facet}) = 169.83 \text{ MPa}$; $\min \sigma_n$ over fault DOFs $\approx 2.15 \text{ MPa}$; $\max \mu_{app} \approx 0.2931 < f_0 \Rightarrow$ **no t=0 pre-slip**.
4. **Nucleation gate.** Compute the barrier $S_E = a \cdot \sigma_n \cdot \ln(V_{dyn}/V_{init}) = 0.015 \cdot 169.83e6 \cdot 27.631 = 70.39 \text{ MPa}$ at the hypocentre facet. The 75 MPa peak clears it by $\times 1.065$ **only** — and the static excess $(f_0 - \mu_{app}) \cdot \sigma_n = 72.11 \text{ MPa}$ gives $\times 1.040$ (**binding**). **These margins are thin** (the deck header itself flags it). If MFEM does not nucleate, raise `delta_tau_pa` to $85.0e6$ ($\times 1.21 / \times 1.18$) *before* touching R or the hypocentre. Quantify G10 here: report $\max |r_{MFEM} - r_{3D}|$ over bump-support DOFs; if it exceeds $\sim 1\%$ of R, the effective bump is measurably wider than SeisSol's.
5. **np=4 smoke, 200 steps, local.** Watch the R-020 `rate_state` at `np>1` warning (G11) — it fires on every SAFS RS run today; confirm it is the ADER-substep path and not the RK path, or accept it.
6. **Short Expanse pilot (user approval required).** `--tfinal 5s, 2N $\times$ 100r, --paraview-fault-dt 0.5s`. Confirm: rupture nucleates in the bump, no NaN, no free-surface daylighting blow-up, V_{max} peaks and decays. Wall-time per simulated second $\times 150$ gives the real budget.
6b. **p2 convergence window (D2).** Same config with `[mesh].order = 2, tfinal = "20s"`. Compare the rupture-front position and peak slip rate against the p1 run at $t \in \{4, 8, 12, 16, 20\} \text{ s}$. Agreement within the snapshot cadence $\Rightarrow$ p1 resolves the cohesive zone; disagreement $\Rightarrow$ report the p1 150 s result as qualitative only. $\sim 4\times$ the p1 cost per simulated second.
7. **Production, checkpoint-chained.** Run the **parity arm** (`sigma_n_strength_floor_pa = -1.0`) first. Inspect the first two fault snapshots for the trace-cell runaway signature before committing the full wall-time; switch to the `snfloor10` arm only if it appears. Compare against `~/Downloads/SAFS_V3.4.1..._outputs/`: fault-surface SRs/SRd/SlS/SlD/Ts0/Td0/Pn0 at matched $t \in \{4, 8, \dots, 96\} \text{ s}$. Reference data stops at $t \approx 98 \text{ s}$. Key metrics: rupture-front arrival along the corridor arclength s , peak slip rate, whether the front crosses the San Gorgonio gate (PREFERRED corridor `kbar_min = 0.98–0.99`, barely above the 0.9 floor — the run decides), final slip, M_w .

Use `safs/seis0l_quakeworx/toolbox/combined_workflow/postprocess.ipynb` for the SeisSol side — it already reads any SeisSol output directory and emits `Mw / slip / stress-drop / finished?`.

1.13 Risks

Risk	Mitigation
Nucleation margins are $\times 1.04$ (binding). MFEM's p1 discretization resolves the 583 m supra-S_E core with ~ 1.5 median fault edges; the bump may fail to nucleate where SeisSol's p3 succeeded. Trace-cell runaway with σ_n strength floor pa = -1 (D1). GTS cost. MFEM has no LTS; SeisSol's $30.5 \times$ LTS speedup is unavailable. Mixed flux is not SeisSol's flux. Central-on-fault-adjacent faces vs pure Godunov. Differences in the rupture front are the <i>point</i> of the experiment, not a bug. p1 vs p3. Under-resolved cohesive zone $\Rightarrow$ artificially high peak slip rate / delayed front. Doc rot. CLAUDE.md and spatial_friction_config_schema.md both make stale claims about spatial-driver nucleation. An implementer following them will do the wrong thing.	Gate 4. Escalate delta_tau_pa $\rightarrow 85e6$ per the deck's own guidance. Run the 10e6 arm in parallel; project_pre_k25_case2_runaway_slip_not_mesh documents the signature (isolated cells sliding ~ 4 m/s forever at σ_n 0.1–5 MPa). Phase 1 measures it. Checkpoint-chain the production run. The mesh is <i>smaller</i> than what MFEM already runs at $2N \times 100r$, so this is a budget problem, not a feasibility one. Also run the <code>--mixed-flux none</code> <code>--time-integrator ader</code> arm (the upwind sibling sbatch) so the flux is the only variable. D2; p2 convergence check on a 20 s window. Phase 3 fixes both.

1.14 Summary of files

Create (6):

safs/project_7.0_preferred/friction/code/build_pref_friction_sidecar.py
uses the shared velocity/code/sidecar.py::write_sidecar; no hand-rolled HDF5

safs/project_7.0_preferred/config/
spatial_friction_safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2.toml
parity arm, σ_n strength floor pa = -1.0
spatial_friction_safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2_snfloor10.toml
fallback arm; differs only in that key + output_dir

seas_test_spatial_friction_sidecar/ 3 tests
+ the f_w = 0 parser/psi tests from Phase 2

jobs/safs/safs_expanse/

spatial_dyn_ratestate_v3_4_1_pref_thermal_case2..._2hr_safs.sbatch (+ restart
↪ sibling)

Change (6):

spatial/code/spatial_friction.hpp	FrictionSidecarSpec, RateStateSidecarFields, resolver ctor
spatial/code/spatial_friction.cpp	:713 f_w >= 0 guard; parse + resolve the sidecar
drivers/spatial_dyn_driver.cpp	load sidecar fields, extend --print-derived
safts/project_7.0_preferred/velocity/code/build_pref_velocity_sidecar.py	
	argparse; stale hard-coded path
safts/project_7.0_alternative/document/spatial_friction_config_schema.md	
	new block; fix stale nucleation claims
miniapps/seas/CLAUDE.md	fix stale "single nucleation kind" paragraph

Re-run, no change (2): - csm_stress_nc_to_mfem_hdf5.py → the Andersonian k=1.8 sidecar -
msh_to_puml.py — not needed; the mesh already exists on both sides

Zero work: mesh, boundary attrs, flux/integrator wiring, FL=103 friction law, compact-bell nucle-
ation, stress reader, material reader, checkpoint/restart, ParaView fault + free-surface output.